AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM351
Data Management and Analysis

Final Examination
Spring Semester 2025/2026

Date: 2 / June / 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

- Total Marks **100** marks
- This exam consists of **Two parts**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

Please select the best and most appropriate answer and write it in the answer booklet.

No.	Question	A	B	C	D
1.	Which type of SQL JOIN returns only the rows where there is a match in both tables?	INNER JOIN	LEFT JOIN	RIGHT JOIN	CROSS JOIN
2.	Which of the following best describes an OLAP system in a data warehouse environment?	It processes day-to-day transactions	It supports complex analytical queries on large datasets	It replaces the need for a relational database	It stores data only in NoSQL format
3.	In a data warehouse, ____ measures can be aggregated along all dimensions using SUM.	Non-additive	Additive	Semi-additive	Derived
4.	Which SQL clause is used to filter results after a GROUP BY operation?	WHERE	HAVING	ORDER BY	GROUP BY
5.	Which of the following is NOT typically a step in the data cleaning process?	Removing duplicate records	Handling missing values	Encrypting all database tables	Correcting inconsistent data formats
6.	Which of the following descriptive analytics techniques groups similar data points together without using labelled training data?	Supervised learning	Regression	Clustering	Decision tree
7.	A relation is in Second Normal Form (2NF) if it is in 1NF and every non-key attribute is _____ on the primary key.	Partially dependent	Fully dependent	Transitively dependent	Independently stored
8.	Which type of NoSQL database stores data as JSON-like documents and is best suited for flexible, schema-less data?	Key-value store	Document store	Column-family store	Graph database
9.	In SQL, a _____ subquery returns exactly one column and one row, and can be used as a single value expression.	Row	Table	Scalar	Correlated
10.	Spreadsheets are easily shared outside the organisation, which creates a problem of data _____.	Cleansing	Sharing	Control	Acquisition

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Part 2: Practical and Theoretical Questions [90 marks]

Question Two: [20 Marks] — SQL

A database stores information about a university: Student, Lecturer, Course, Enrollment, Exam, Grade, and Scholarship tables.

Schema:

- Department (**Dept_ID** [PK], Dept_Name, Location, Head_Lecturer_ID)
- Student (**Student_ID** [PK], First_Name, Last_Name, Gender, Date_of_Birth, Email, Phone, Major, Dept_ID)
- Lecturer (**Lecturer_ID** [PK], First_Name, Last_Name, Specialization, Email, Phone, Dept_ID)
- Course (**Course_ID** [PK], Course_Name, Credits, Start_Date, End_Date, Dept_ID , Lecturer_ID)
- Enrollment (**Enrollment_ID** [PK], Student_ID , Course_ID , Enrollment_Date, Status)
- Exam (**Exam_ID** [PK], Course_ID , Exam_Date, Exam_Type, Total_Marks)
- Grade (**Grade_ID** [PK], Student_ID , Exam_ID , Marks, Remarks)
- Scholarship (**Scholarship_ID** [PK], Student_ID , Scholarship_Name, Amount, Award_Date, Status)

Write SQL statements for the following:

A. List the first name, last name and major of all students who enrolled after 1 January 2025. ***[4 marks]***

B. Find lecturers who teach more than two courses. List their first names and the number of courses they teach. ***[3 marks]***

C. List all course names and the number of students enrolled in each course, ordered by the number of enrollments in descending order. ***[2 marks]***

D. Update the status of all scholarships to 'Closed' where the award date is before 1 January 2024 and the current status is 'Active'. ***[4 marks]***

E. Calculate the average marks obtained and the number of students for each course in the Grade table. Include only courses where the average marks exceed 60. ***[3marks]***

F. Delete all scholarship records for students whose major is 'Business Administration'. (Hint: use subquery) ***[4 marks]***

Question Three: [20 Marks] — MongoDB

Use the sample below from the 'order' document collection (database: "db", collection: "order"):

```
{ "order_id": "ORD001", "customer_name": "Ali Al Rashdi",
  "order_date": "2025-03-10", "status": "Delivered", "total_amount": 245.50,
  "payment_method": "Credit Card", "delivered": true,
  "items": [
    { "product_name": "Laptop", "quantity": 1, "price": 180.00 },
    { "product_name": "Mouse", "quantity": 2, "price": 25.00 },
    { "product_name": "USB Hub", "quantity": 1, "price": 15.50 },
    { "product_name": "Keyboard", "quantity": 1, "price": 25.00 }
  ]
}
```

PROJECT

A. Display all documents in the order collection. ***[2 marks]***
B. Display all orders with a status of 'Delivered'. ***[3 marks]***
C. Display the first four orders where the total amount is greater than 200. ***[4 marks]***
D. Find all orders that contain at least one item with a price greater than 100. ***[5 marks]***
E. Find all orders that were paid by 'Credit Card' and have been delivered. ***[6 marks]***

Question Four: [50 Marks] — Theory & Data Analysis

A. Define the term 'subquery' in SQL. Discuss two different ways subqueries can be used within an SQL statement. Explain what a correlated subquery is and how it differs from a non-correlated subquery. ***[6 marks]***

B. How can indexes improve the performance of a search query in SQL? Explain the concept of an index and describe a scenario where creating an index on a column significantly improves query performance. ***[6 marks]***

C. Explain the roll-up and drill-down operations in a data warehouse. How do they make use of dimensional hierarchies and measures to perform aggregations? Give one example of each. ***[6 marks]***

D. Can normalization avoid 100% redundancy in a relational database? Justify your answer with a clear explanation and at least one example. ***[6 marks]***

E. An experiment involves two groups of participants. The scores of each group member for two subjects, Physics and Chemistry, are:

1. Group A: A1 — Physics: 82, Chemistry: 78
2. Group B: B1 — Physics: 65, Chemistry: 70

A new student S1 has Physics score = 74 and Chemistry score = 72. Calculate the similarity ratio of S1 with A1 and B1. ***[10 marks]***

F. In a silhouette analysis, point Pk is in cluster M. The average distances from Pk to each cluster are: Cluster M = 8, Cluster N = 20, Cluster O = 35. Calculate the silhouette coefficient of Pk. Explain each step clearly. ***[10 marks]***

G. What is the difference between structured, semi-structured, and unstructured data? Give one real-world example of each type and explain which is most suitable for storage in a relational database. ***[6 marks]***

End of the questions

EXCERPT